Module 3 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 3, Quiz B: Concept check

Reading: Module 3 Read 2 (complete Read 1 + Quiz A first)
Canvas: Concept Check quiz (Module 3)

Apply graph selection, Pearson r, association description, two-way table hand proportions, and graph image items from Read 2 (Sections 4–5: two categorical, two numerical).

Full integration and demo-lab R patterns → Synthesis.

Section A — Graphs: identify and interpret (Read 2 §2–3 + Read 1)

A1. Which graph for numerical by group?

You have a numerical outcome (`body_mass_g`) measured for three species (categorical). Which graph best compares the distributions?

A2. Which graph for two numerical?

You have two numerical variables: `bill_length_mm` (x) and `flipper_length_mm` (y). Which graph?

A3. Which graph for two categorical?

You have two categorical variables: `species` and `island`. Which graph shows how they co-occur?

A4. Read a boxplot

A boxplot has lower fence 10, $Q_1 = 18$, median = 25, $Q_3 = 32$, upper fence 45. What is the IQR?

A5. Skew from boxplot

A boxplot shows: min=5, $Q_1 = 10$, median=12, $Q_3 = 20$, max=50. Is the distribution skewed? Which way?

Answer key — Section A — Graphs: identify and interpret (Read 2 §2–3 + Read 1)

1. Boxplot per group (side-by-side boxplots with `group` on x-axis) — or grouped histograms/dot plots. Boxplots clearly show median, spread, and outliers per group.
2. A scatterplot (bill length on the x-axis, flipper length on the y-axis).
3. A stacked bar chart (or 100% stacked bar) showing the species composition within each island.
4. IQR $= Q_3 - Q_1 = 32 - 18 = 14$.
5. Right-skewed (positively skewed) — the upper whisker (12 to 50) is much longer than the lower whisker, and the median is closer to Q_1 than Q_3.

Section B — Choosing the right summary by variable type

B1. Two numerical variables

Both variables are numerical (`height` and `weight`). What is the first graph to make? What do you describe?

B2. Two categorical variables

Both variables are categorical (`diet_type` with 3 levels, `outcome` with 2 levels). Best graph and summary?

B3. Numerical + categorical

One variable is numerical (`blood_pressure`) and one is categorical (`treatment_group`: A/B/C). Best graph?

B4. Univariate: one categorical

One variable: `habitat` with levels Urban/Suburban/Rural. Best graph and summary?

Answer key — Section B — Choosing the right summary by variable type

1. Scatterplot. Describe form (linear/curved/none), direction (positive/negative/none), strength (strong/moderate/weak).
2. Stacked bar chart (counts or 100% proportions). Two-way table of counts and row/column proportions.
3. Side-by-side boxplots (treatment group on the x-axis) or grouped histograms. Summarize center and spread within each group.
4. Bar chart. Summary: counts and proportions per level.

Section C — Association description and two-way tables (Read 2)

C1. Positive association

Describe a positive association between two numerical variables in words.

C2. Form vs direction

What is the difference between form and direction in a scatterplot description?

C3. Row proportion

Two-way table: Robin Urban=8, Robin Rural=2 (row total=10). Sparrow Urban=4, Sparrow Rural=6 (row total=10). What is the row proportion of Sparrows that are Rural?

C4. Column proportion

Two-way count table (species × habitat): Robin Urban=8, Robin Rural=2. Sparrow Urban=4, Sparrow Rural=6. Column totals: Urban=12, Rural=8. What is the column proportion of Urban birds that are Robin?

C5. Joint vs marginal

What is the difference between a joint proportion and a marginal proportion in a two-way table?

C6. 100% stacked bar

When would you prefer a 100% stacked bar (each bar scaled to 100%) over a regular stacked (count) bar?

Answer key — Section C — Association description and two-way tables (Read 2)

1. As one variable increases, the other tends to increase as well (upward trend in the scatterplot).
2. Form is the shape of the trend (linear, curved, no pattern). Direction is whether higher x goes with higher y (positive), lower y (negative), or no consistent change (none).
3. $6/10 = 0.60$ — 60% of Sparrows are in Rural habitat (denominator is Sparrow row total).
4. $8/12 \approx 0.667$ — about 67% of Urban birds are Robins (denominator is Urban column total).
5. Joint: fraction of ALL observations in one specific cell (divide by grand total). Marginal: fraction in a row or column total (divide by grand total for that row or column).
6. When you want to compare proportions within groups rather than raw counts — especially when group sizes differ, so the bars all reach 100% and the composition is easier to compare.

Section D — Scatterplot interpretation (3)

D1. Positive linear

Hours studied vs exam score: points rise in a straight band. Describe form, direction, strength.

D2. Negative linear

Screen time vs sleep: higher screen pairs with lower sleep, loosely scattered. Describe association.

D3. No pattern

Shoe size vs quiz score: cloud with no slope. Describe association.

D4. Curved caution

Scatterplot shows a U-shape. Can you summarize with Pearson r alone?

Answer key — Section D — Scatterplot interpretation (3)

1. Form: linear. Direction: positive. Strength: moderate to strong.
2. Form: linear. Direction: negative. Strength: weak to moderate.
3. Form: none. Direction: none. Strength: none (no association).
4. No — r measures linear association; describe curved form first.

Section E — Pearson r interpretation (3)

E1. Strong positive r

$r = 0.92$ for bill length vs flipper length. Interpret.

E2. Moderate negative r

$r = -0.55$ for advertising spend vs returns. Interpret.

E3. Weak r

$r = 0.12$ for height vs test score. Interpret.

E4. r magnitude

Which $|r|$ is stronger: -0.88 or $+0.45$?

Answer key — Section E — Pearson r interpretation (3)

1. Strong positive linear association — longer bills tend to pair with longer flippers.
2. Moderate negative linear association — higher spend tends to pair with lower returns.
3. Weak positive linear association — little linear trend.
4. $|-0.88| = 0.88$ is stronger — sign is direction, magnitude is strength.

Section F — Hand computation: two-way proportions (3 datasets)

F1. Row proportion

Two-way table (species $\times$ habitat, $n = 20$): Sparrow in Rural = 6, Sparrow row total = 10. What is the row proportion of Sparrows that are Rural?

F2. Column proportion

Two-way table (species $\times$ habitat): Robin in Urban = 8, Urban column total = 12. What is the column proportion of Urban birds that are Robin?

F3. Joint proportion

Two-way table (species $\times$ habitat, grand total $n = 20$): Robin in Urban = 8. What is the joint proportion that are Robin AND Urban?

F4. Treatment table

Two-way table (treatment $\times$ outcome, $n = 50$): Treatment B successes = 10, Treatment B row total = 25. What is the row proportion of successes within Treatment B?

F5. Island table

Two-way table (species $\times$ island): Gentoo on Biscoe = 9, Gentoo row total = 10. What is the row proportion of Gentoo penguins that are on Biscoe?

Answer key — Section F — Hand computation: two-way proportions (3 datasets)

1. $6/10 = 0.60$.
2. $8/12 \approx 0.667$.
3. $8/20 = 0.40$.
4. $10/25 = 0.40$.
5. $9/10 = 0.90$.

Section G — Plot types: scatter, stacked bars, side-by-side boxplots (verbal)

G1. Scatterplot variables

You have two numerical variables. What is the first graph to make?

G2. Stacked bar counts

You have two categorical variables. Which graph shows raw counts of one variable stacked within each level of the other?

G3. 100% stacked bar

You have two categorical variables and want to compare composition (proportions) across groups when the group sizes differ. Which graph?

G4. Grouped boxplots

Blood pressure (numerical) measured for three treatments (A/B/C, categorical). Best graph?

G5. Two-way table first

You have two categorical variables. Before computing row proportions, what table do you build?

G6. Color scatter by group

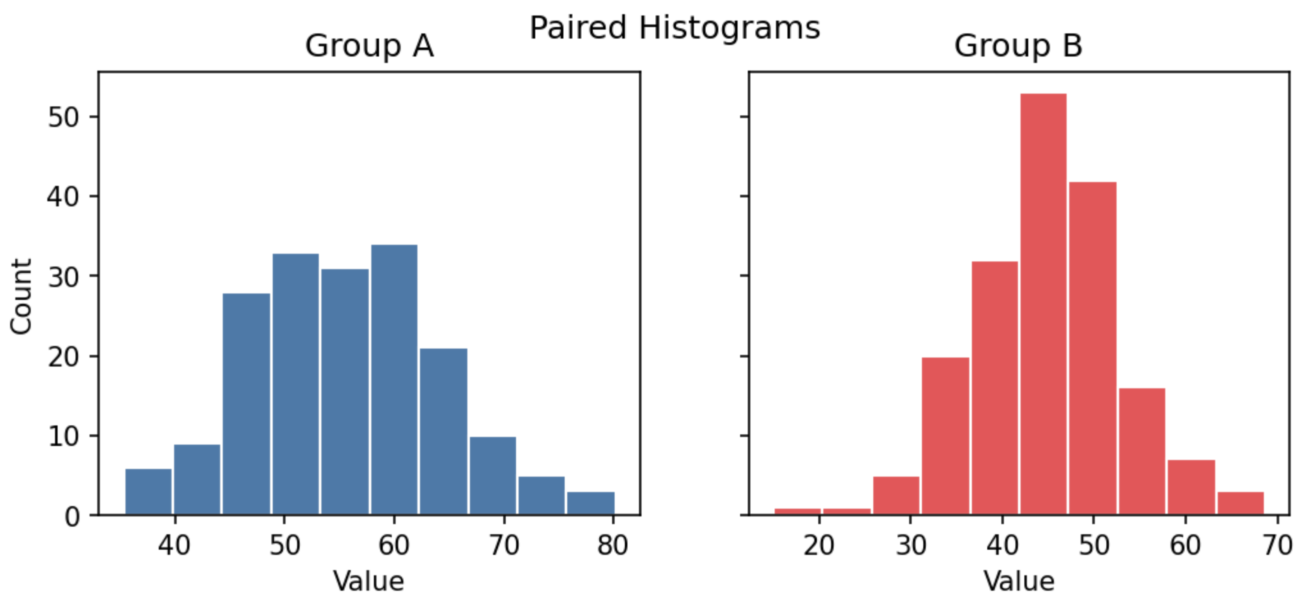
You make a scatterplot of two numerical variables and color the points by a categorical variable (species). What extra pattern might you see?

Answer key — Section G — Plot types: scatter, stacked bars, side-by-side boxplots (verbal)

1. Scatterplot.
2. Stacked bar (raw counts stacked within each group).
3. 100% stacked bar (each bar scaled to 100%).
4. Side-by-side boxplots.
5. Two-way count table.
6. Whether the association differs by species (subgroups).

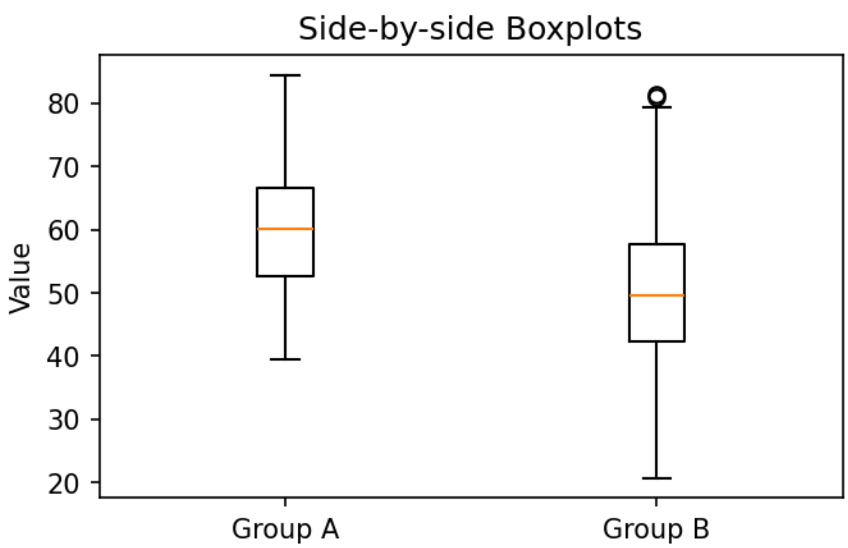
Section H — Graph interpretation (images · Quiz B)

IMG5. Paired histograms



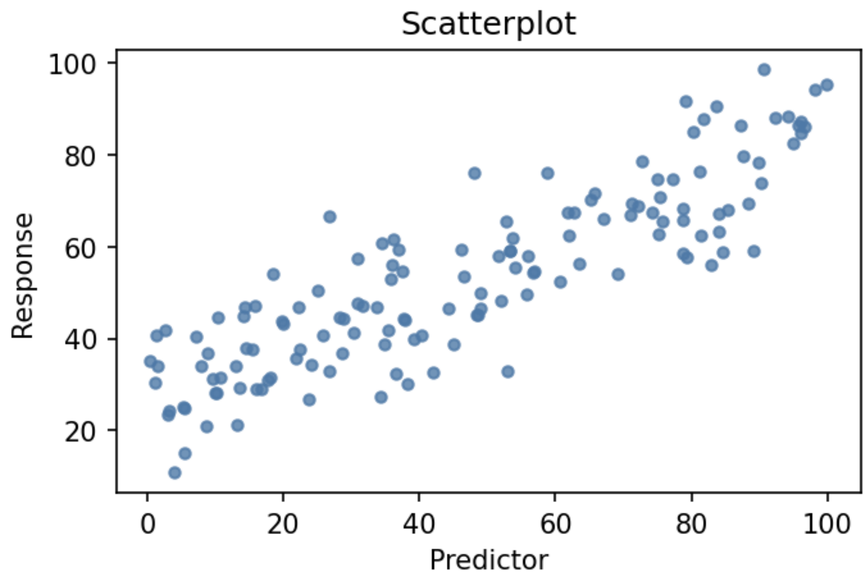
Which group appears to have the higher typical value?

IMG7. Side-by-side boxplots



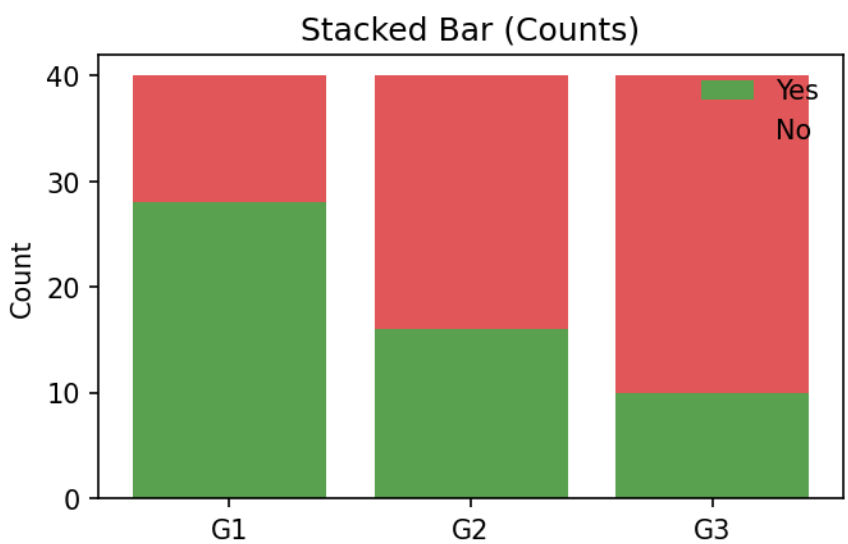
Which group has larger spread (IQR)?

IMG8. Scatterplot form-direction-strength



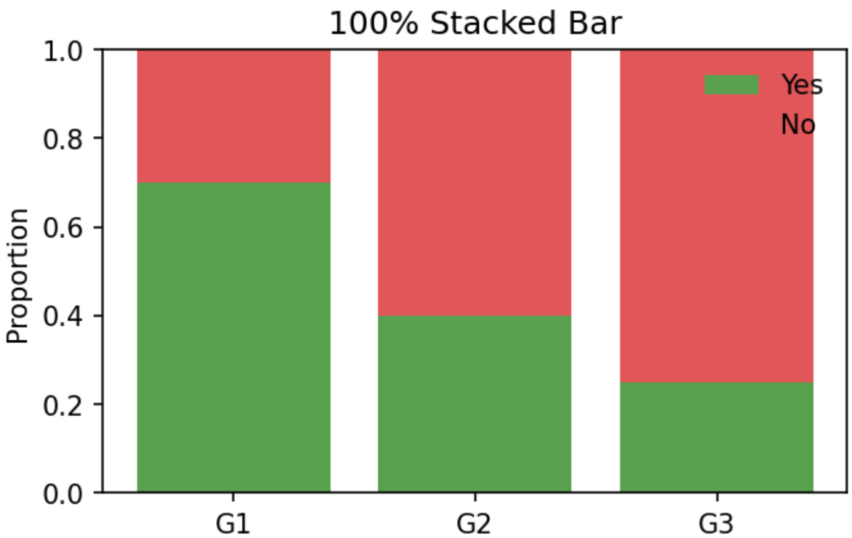
Best description of this scatterplot?

IMG10. Stacked bar counts



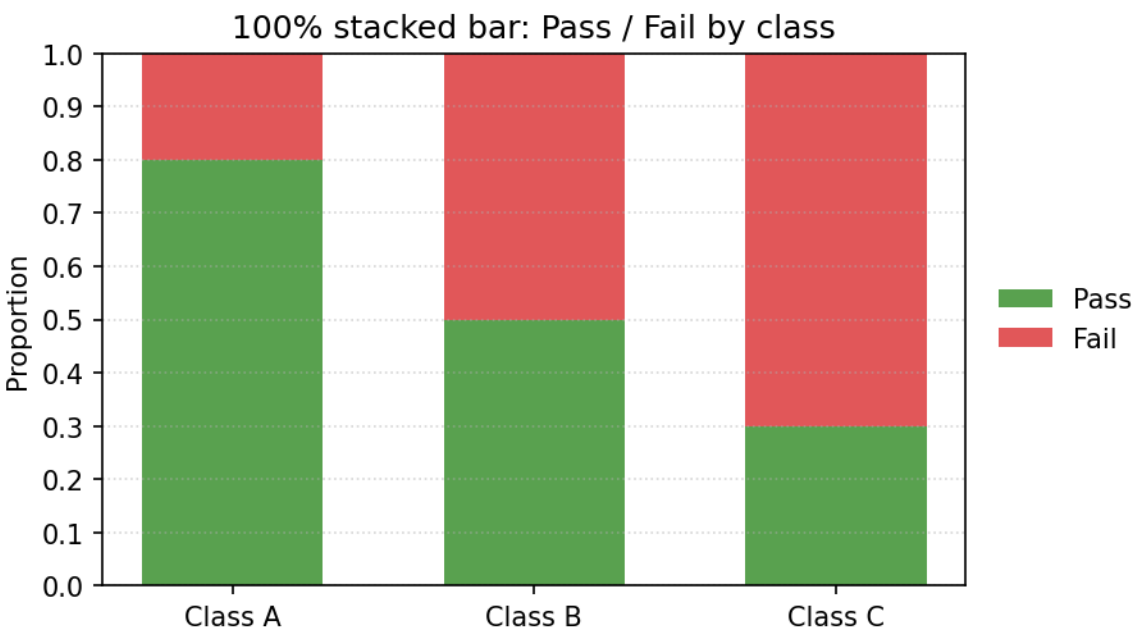
This is a stacked bar chart with raw counts. What is easiest to compare?

IMG11. 100% stacked bar



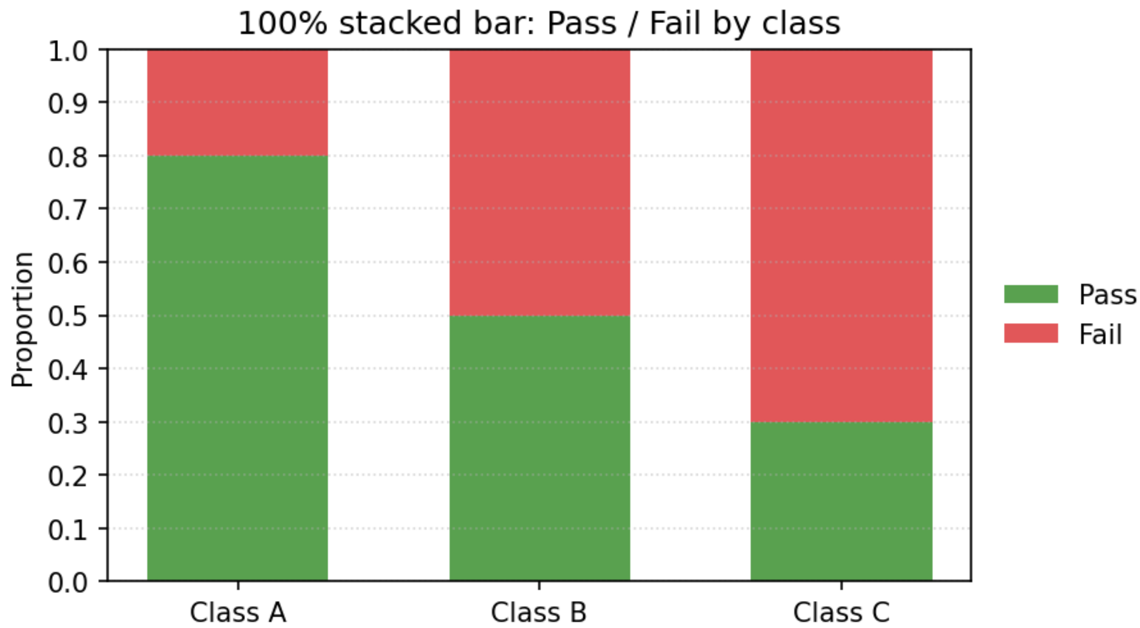
This is a 100% stacked bar chart. What is easiest to compare?

IMG17. 100% stacked bar — proportion Pass in Class A



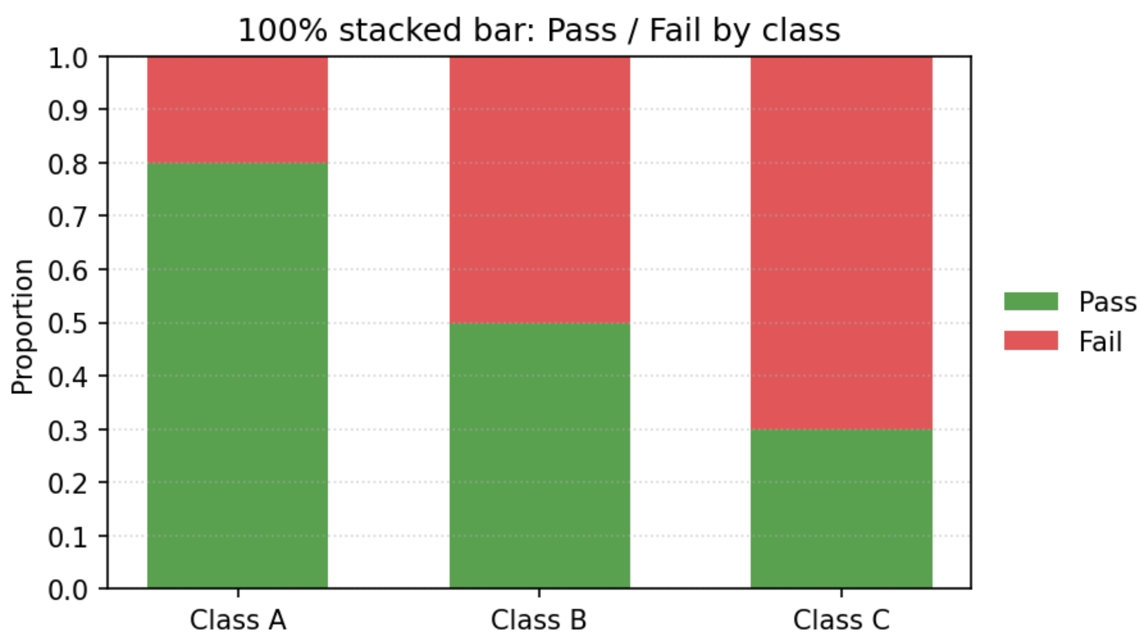
In this 100% stacked bar chart, estimate the proportion of Class A that Passed.

IMG18. 100% stacked bar — proportion Fail in Class C



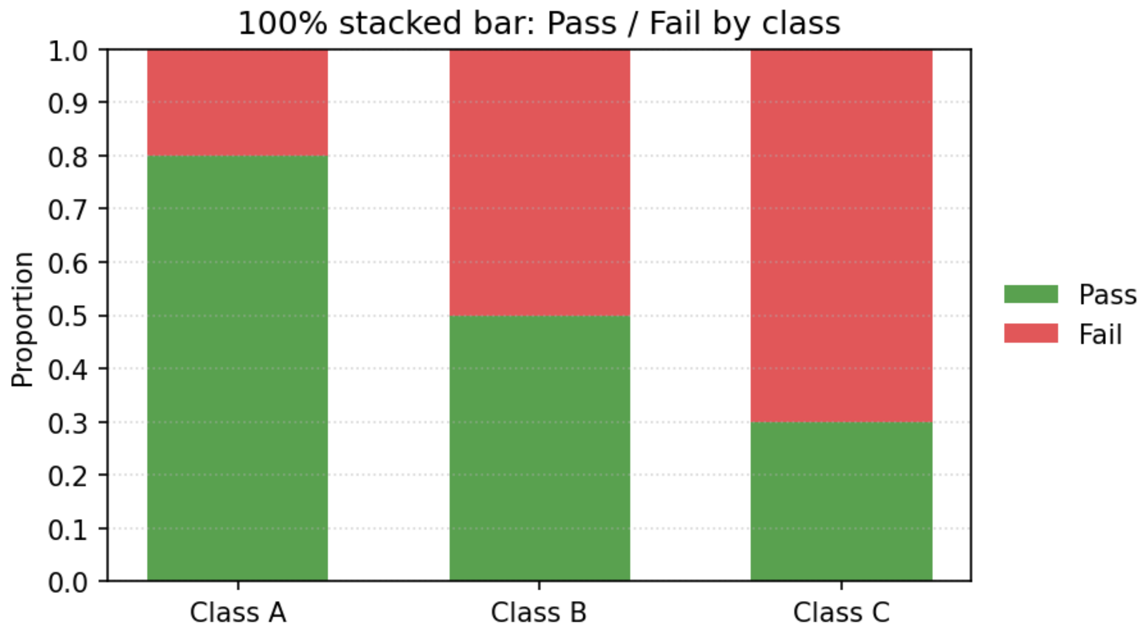
In this 100% stacked bar chart, estimate the proportion of Class C that Failed.

IMG19. 100% stacked bar — highest Pass proportion



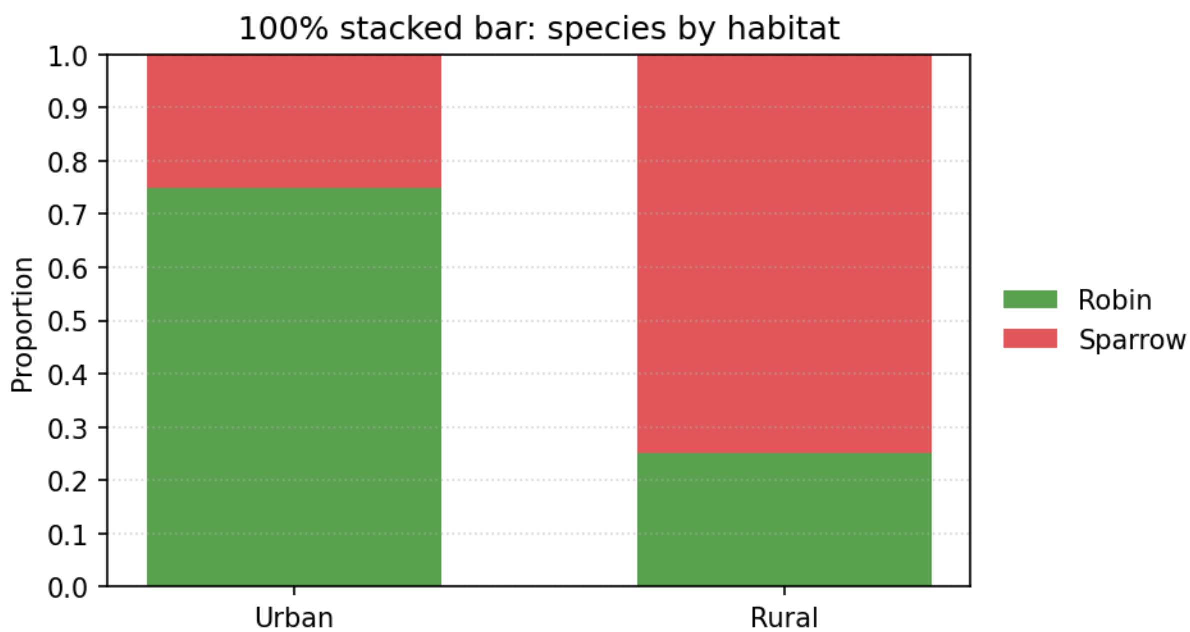
Which class has the highest proportion of students who Passed?

IMG20. 100% stacked bar — proportion Pass in Class B



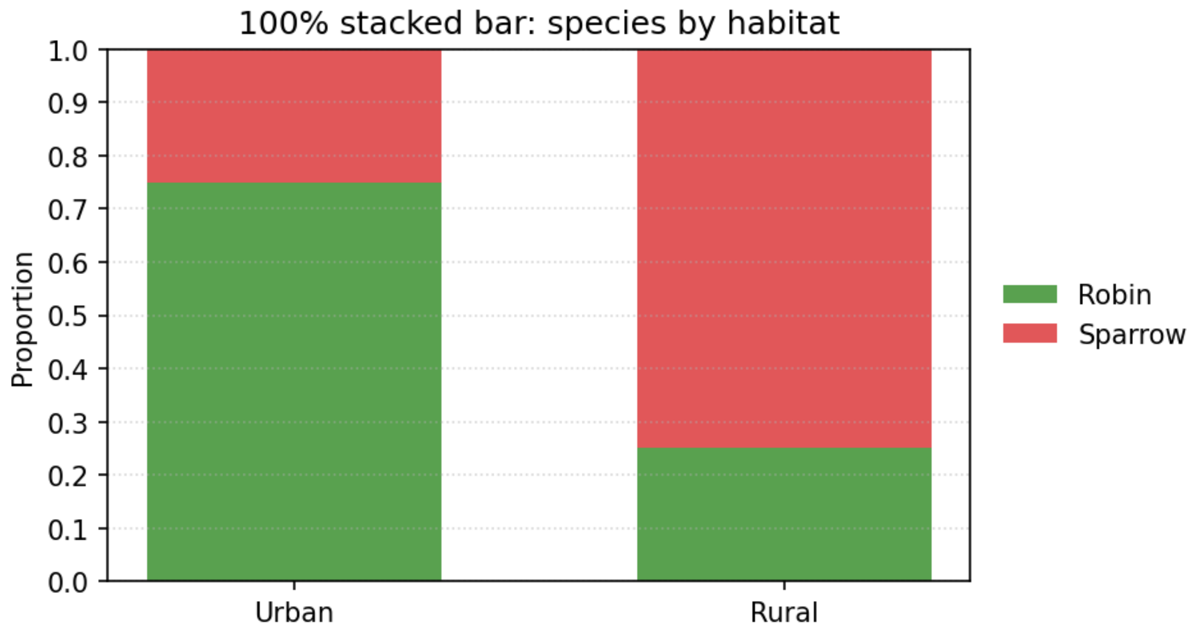
In this 100% stacked bar chart, estimate the proportion of Class B that Passed.

IMG21. 100% stacked bar — proportion Robin in Urban



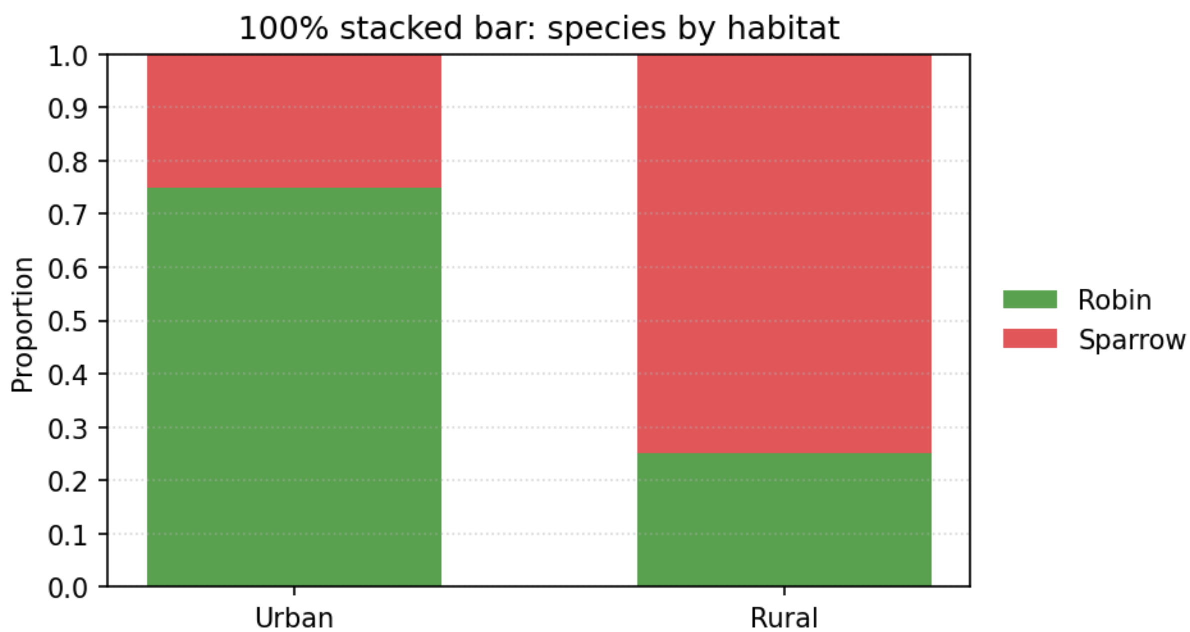
In this 100% stacked bar chart, estimate the proportion of Urban birds that are Robins.

IMG22. 100% stacked bar — proportion Sparrow in Rural



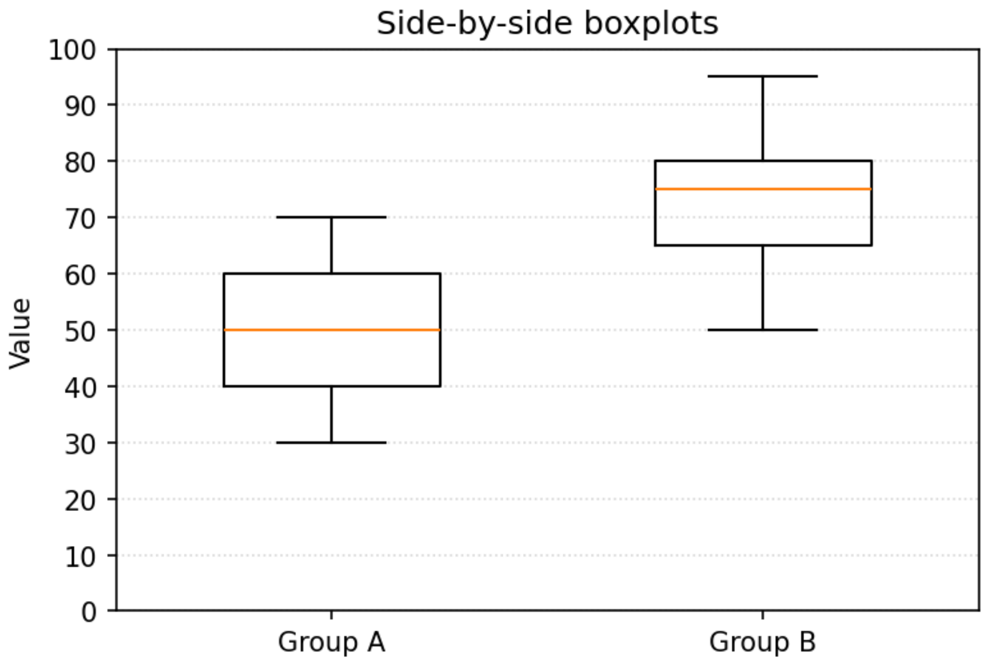
In this 100% stacked bar chart, estimate the proportion of Rural birds that are Sparrows.

IMG23. 100% stacked bar — higher Sparrow proportion



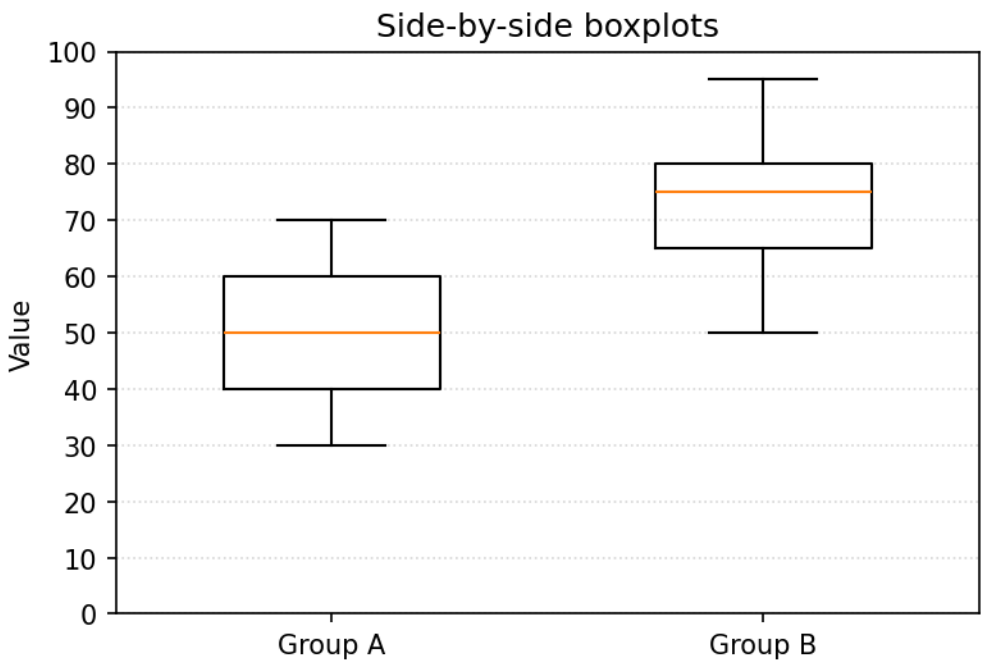
Which habitat has the higher proportion of Sparrows?

IMG24. Side-by-side boxplots — higher median



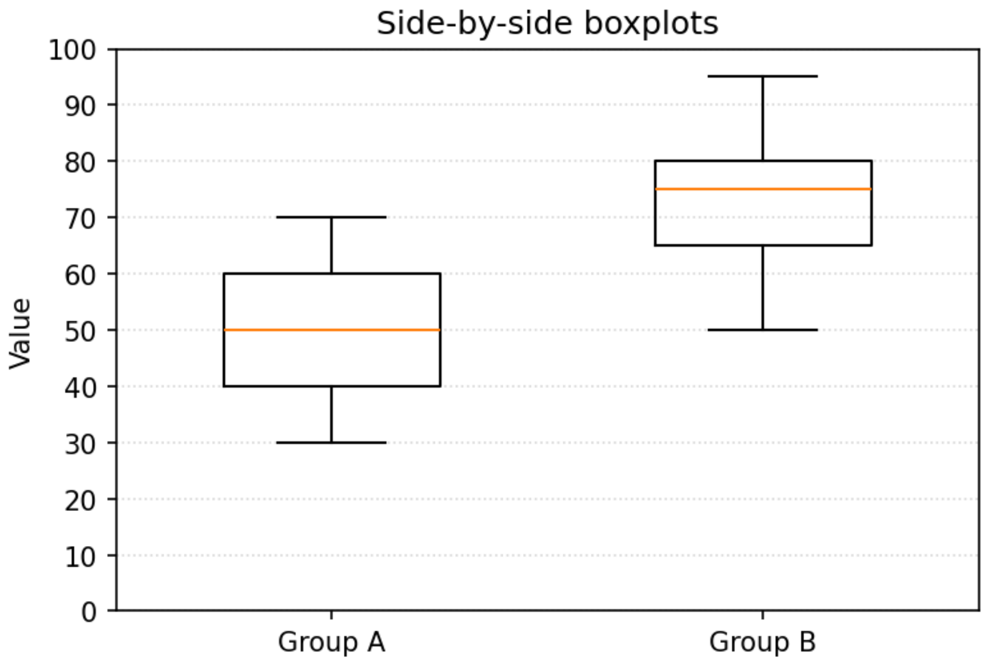
Which group has the higher median?

IMG25. Side-by-side boxplots — more spread



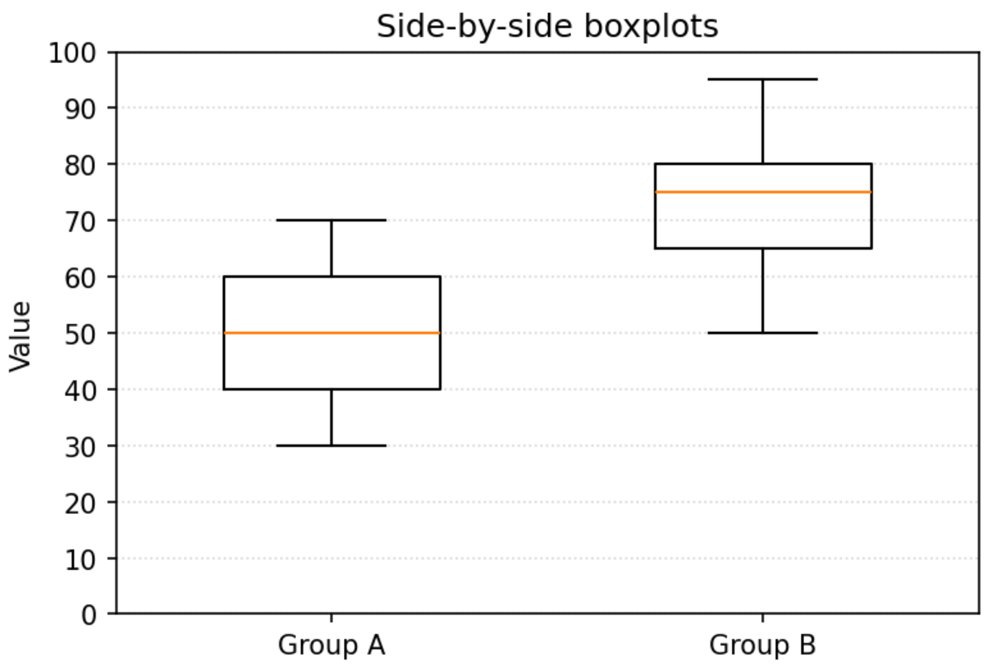
Which group has more spread (larger IQR)?

IMG26. Side-by-side boxplots — estimate median



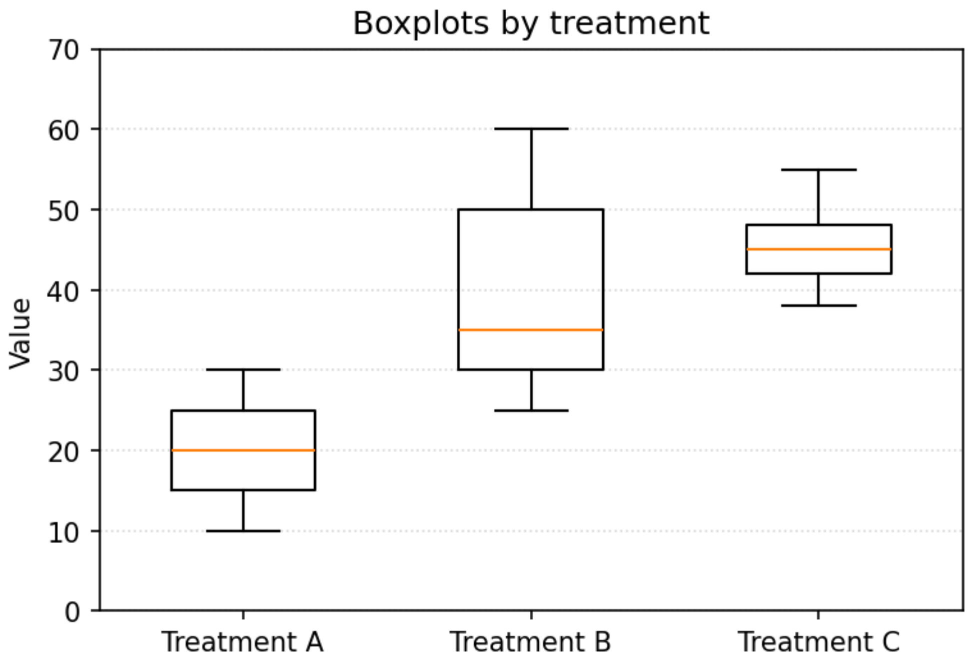
Estimate the median of Group A (the line inside its box).

IMG27. Side-by-side boxplots — estimate IQR



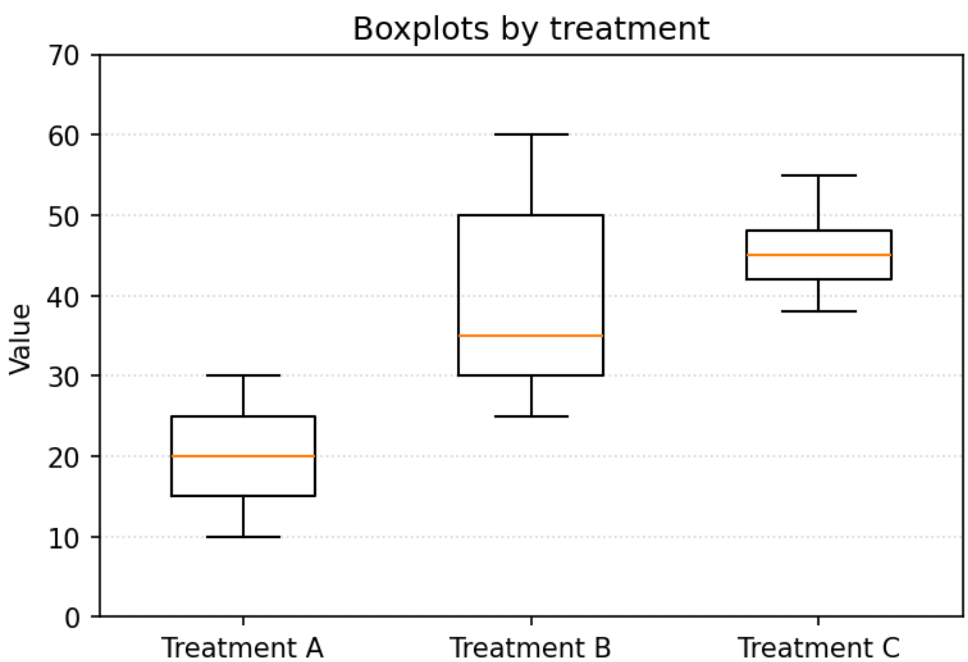
Estimate the IQR (box height) of Group B.

IMG28. Three boxplots — highest median



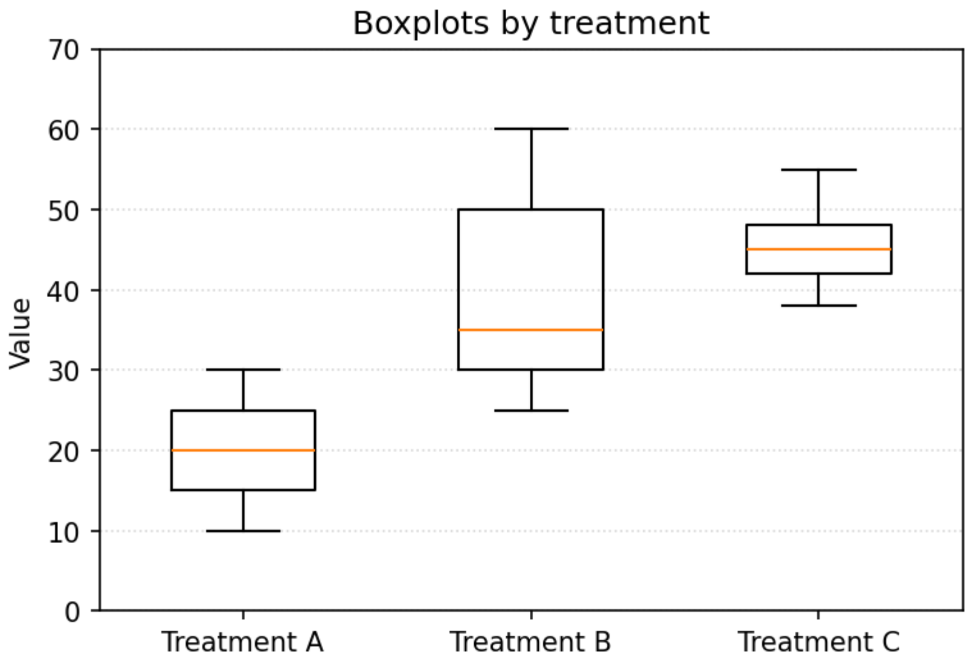
Which treatment has the highest median?

IMG29. Three boxplots — largest spread



Which treatment has the largest spread (IQR)?

IMG30. Three boxplots — smallest spread



Which treatment has the smallest spread (IQR)?

Answer key — Section H — Graph interpretation (images · Quiz B)

1. Group A (its distribution is shifted higher).
2. Group B (wider box / longer whiskers).
3. Roughly linear, positive direction, moderate-to-strong association.
4. Total counts by group.
5. Within-group proportions across groups.
6. About 0.80 — the Pass segment fills roughly 80% of Class A's bar.
7. About 0.70 — the Fail segment fills roughly 70% of Class C's bar.
8. Class A (about 80% Pass, the tallest Pass segment).
9. About 0.50 — the Pass and Fail segments are about equal for Class B.
10. About 0.75 — the Robin segment fills roughly three-quarters of the Urban bar.
11. About 0.75 — the Sparrow segment fills roughly three-quarters of the Rural bar.
12. Rural (about 75% Sparrow vs about 25% in Urban).
13. Group B (median 75 vs 50 for Group A).
14. Group A (IQR 20, box from 40 to 60) vs Group B (IQR 15).
15. About 50.
16. About 15 — the box runs from about Q1 = 65 to about Q3 = 80.
17. Treatment C (median 45, vs 35 for B and 20 for A).
18. Treatment B (IQR 20, box from 30 to 50).
19. Treatment C (IQR 6, the narrowest box).